

Aircraft Engine Remaining Useful Life Prediction

Project Proposal for Parallel Processing of Large Sensor Data

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Research objective

Predict remaining useful life (RUL), measured in flight cycles until failure, from NASA N-CMAPSS DS02. Build a reproducible analysis and inference pipeline that compares regression on statistical features with a compact 1D CNN, while measuring reading, feature-extraction and parallel-processing costs.

Data and prediction methods

DS02 has 6,517,190 rows from nine simulated engines [1, 2]. Inputs contain only 18 operating-condition and measured-sensor channels (W and Xs), excluding degradation variables T, health state hs and virtual sensors Xv. Engine and flight-cycle boundaries are preserved. Mean, standard deviation, minimum, maximum and slope provide 90 features for a mean baseline, Ridge and histogram gradient boosting. A 1D CNN uses raw sensor windows. Six window/stride settings and raw/nonnegative prediction policies are also evaluated.

Evaluation and computing experiments

Select settings using three-fold engine-grouped validation on development engines 2, 5, 10, 16, 18 and 20. Average window predictions at flight end, then report cycle-level MAE and RMSE per engine and their unweighted means. Test engines 11, 14 and 15 were inspected in the original experiment; extensions are explicitly exploratory re-evaluations of this same test split. Compare 1, 2, 4 and 8 workers and HDF5 chunk sizes for time, throughput, memory and output agreement. Three simulated test engines do not establish real-aircraft operational performance.

Implementation record and submission dates

Dates in 2026	Work and deliverables
2 October	Data, pipeline, baselines, sensitivity study, CNN, inference and experiments
21 October	Submit separate one-page Korean and English proposals
8 December	Five-minute presentation
16 December 23:59	Submit report and source code

References

[1] NASA PCoE Data Set Repository, Dataset 17, Turbofan Engine Degradation Simulation-2.
[2] Arias Chao et al. (2021). Aircraft Engine Run-to-Failure Dataset under Real Flight Conditions for Prognostics and Diagnostics. Data, 6(1), 5. doi:10.3390/data6010005.
NASA is the original source; DS02 was acquired through the Figshare third-party mirror (doi:10.6084/m9.figshare.20436504.v1). The NASA full bundle is 15.76 GB; the DS02 file is 2.45 GB.